The secret life of clouds: differences in the day – night stratocumulus cloud response to aerosol

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IMPERIAL

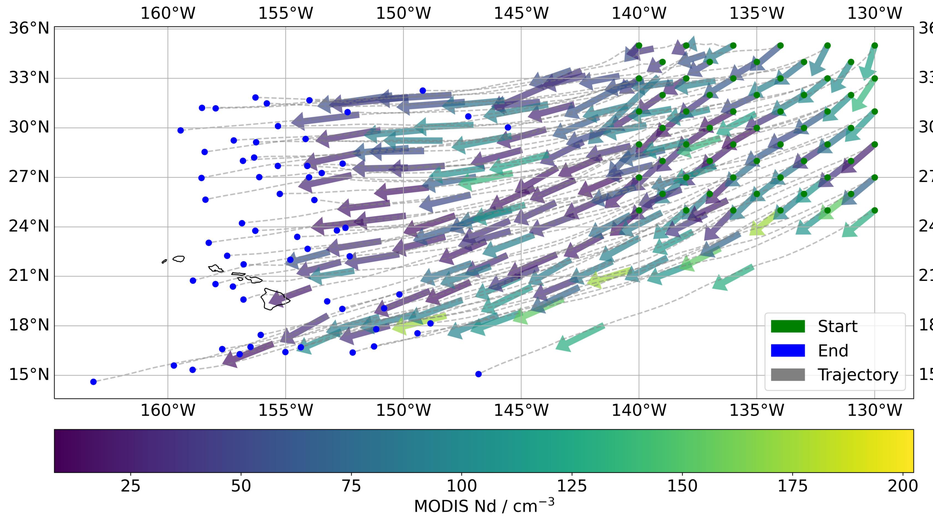
Cloud fraction response to aerosol driven by nighttime processes

Motivation

Previous observational studies into aerosol cloud interactions have focused on the daytime behaviour → Significant observational gap in cloud nighttime behaviour
Stratocumulus clouds have a pronounced diurnal cycle (Wood 2012)
Cloud nighttime processes set the cloud state at the start of the day, which is a strong control on the clouds' development (Gryspeerdt+ 2022)
Here we:

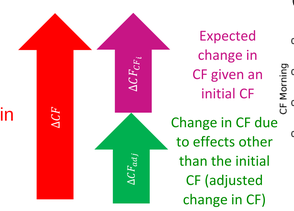
- → Compare the day and night cloud amount response to aerosol
- → Attempt to isolate the aerosol impact, by subtracting off the contributions from the clouds initial state

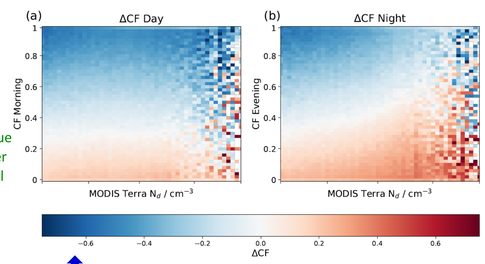
Lagrangian perspective: how does the initial N_d impact the subsequent development?



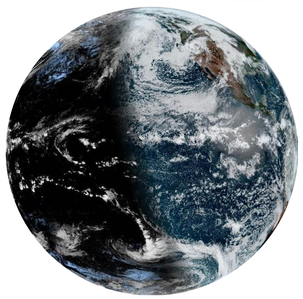
Above: Illustration of the Lagrangian method used. All trajectories are initialized daily at 06:00 LST off the coast of California and advected for three days in the 1000hPa wind field. The trajectories are collocated with the MODIS morning overpass, repeated over a four year period (~ 2 million individual trajectories are analysed)

Remove the impact of the initial cloud fraction on the development





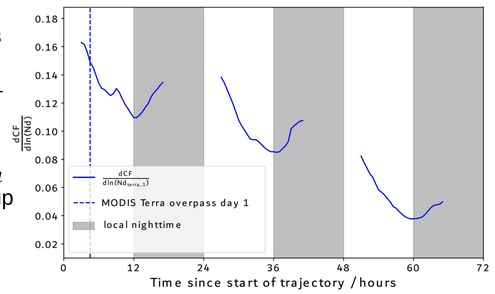
Calculate changes in the CF and LWP over the day and night in bins of initial CF or LWP and N_d



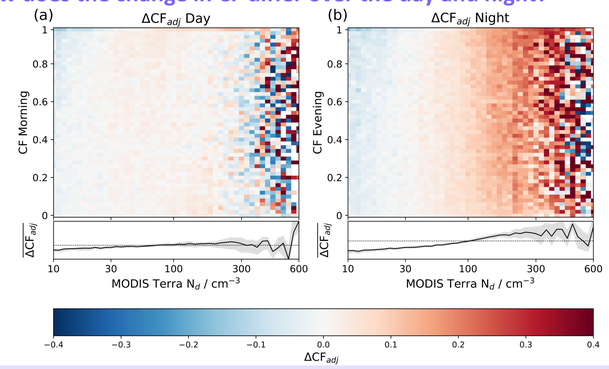
Cloud microphysical retrievals are generally not possible during the nighttime using passive instruments (Nakajima and King) therefore this study considers changes in the cloud state between dusk and dawn

Compare the CF - N_d relationship over the diurnal cycle

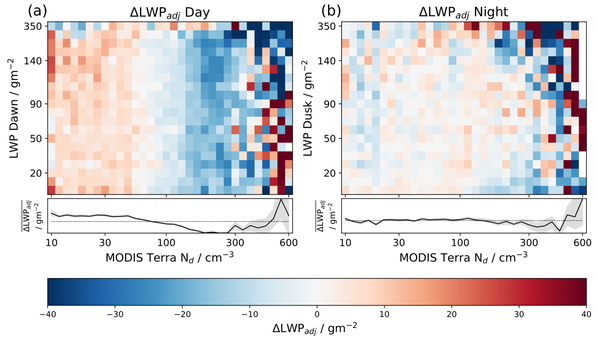
A single snapshot is not sufficient to characterize the CF - N_d sensitivity
→ During the nighttime the CF - N_d relationship builds up and then weakens during the daytime



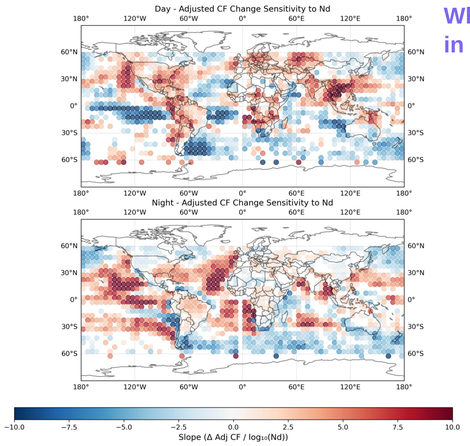
How does the change in CF differ over the day and night?



How does the change in LWP differ?



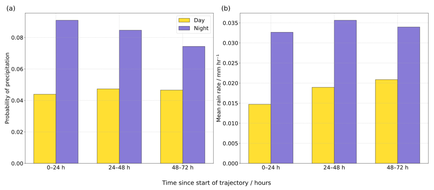
What are the global patterns in this relationship?



Using a similar method as for the Californian stratocumulus deck, the relationship between the adjusted change in CF and N_d is examined

Generally areas associated with higher stratocumulus cloud amount have a more positive $\Delta CF_{adj} - N_d$ relationship during the nighttime

Is this consistent with a precipitation suppression mechanism?



Stratocumuli are more likely to be precipitating during the nighttime

→ Suggests that a precipitation suppression impact would be strongest during this time

References:

Gryspeerdt, E., Glassmeier, F., Feingold, G., Hoffmann, F., and Murray-Watson, R. J.: Observing short-timescale cloud development to constrain aerosol–cloud interactions, Atmospheric Chemistry and Physics, 22, 11 727–11 738, https://doi.org/10.5194/acp-22-11727-2022, 2022.

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There is an observational requirement for cloud nighttime observations:

The CF response to aerosol is driven by nighttime processes

→ This suggests that aerosol play an important role in cloud formation (since it is during the nighttime that stratocumuli typically build up) and a lesser impact on the cloud breakup

→ Entrainment dissipates LWP during the daytime, but precipitation suppression offsets this effect during the nighttime

Further observational work is required to understand the balance of different cloud microphysical processes over the diurnal cycle